

# Xiang He

Ph.D. Candidate in Geosciences, The University of Texas at Austin

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## Research Profile

Computational geodynamicist studying how mantle flow, subduction, lithospheric deformation, and mantle rheology shape observable surface motions. I build physics-based numerical models and compare model predictions with geological and geophysical constraints, with current emphasis on Western U.S. vertical motion and Cascadia.

## Education

### The University of Texas at Austin

*Ph.D. in Geosciences / Geophysics*

**Aug. 2024–Present**

*Austin, TX, USA*

- Advisor: Prof. Thorsten W. Becker
- Research focus: computational geodynamics, mantle flow, subduction, dynamic topography, vertical crustal motion

### University of Science and Technology of China

*B.S. in Geophysics*

**Sep. 2020–Jun. 2024**

*Hefei, Anhui, China*

- Guo Moruo Scholarship; China National Scholarship

## Research Experience

### Western U.S. Vertical Motion — UT Austin

**Aug. 2024–Present**

*Graduate Researcher, UT Institute for Geophysics, Advisor: Thorsten Becker*

*Austin, TX*

- Build global and regional geodynamic models to quantify how subduction, mantle flow, and rheology influence vertical crustal motion and dynamic topography.
- Develop Cascadia subduction-zone model configurations and compare model predictions with geological/geophysical observations.
- Design numerical experiments to test sensitivity to slab geometry, mantle rheology, density structure, and boundary conditions.
- Run and analyze large-scale simulations using ASPECT and high-performance computing workflows; process model output with Python, GMT, and related tools.

### Western Pacific double-subduction drives Gulf of California rifting — UT Austin

**Feb. 2025–Jul. 2026**

*Graduate Researcher, UT Institute for Geophysics, Advisor: Thorsten Becker*

*Austin, TX*

- Set up global reference geodynamic model to evaluate velocity field and stress state of lithosphere.
- Compare double and single subduction configurations, evaluate the additional extensional stress ( 15 MPa) imparted on Gulf of California rifting from western Pacific double-subduction system.

### 3D Regional Model: Intraplate Deformation within East Asia (remote) — UIUC

**May. 2023–Oct. 2023**

*Visiting Student, Department of Earth Science and Environmental Change, Advisor: Lijun Liu*

*Urbana, IL*

- Conducted time-dependent 3-D regional geodynamic modeling of intraplate deformation in East Asia.
- Analyse the velocity field and stress field near Tan-Lu Fault and investigate how its strike-slip pattern changes through time.

### Rock Dynamics Simulation of Planetary Crust Formation — USTC

**Jan. 2023–Aug. 2023**

*Undergraduate Researcher, School of Earth and Space Sciences, Advisor: Wei Leng*

*Hefei, China*

- Set up a 3D regional model to evaluate the effect of hydrated plume on continental crust formation.
- Model shows that the generation of continental crust is almost impossible with a dry plume, which indicates the necessity of deep water's participation in this process.

### Réunion plume and double-subduction coupling as a driver of India's two-stage rapid northward motion — USTC

**Apr. 2022–Jun. 2024**

*Undergraduate Researcher, School of Earth and Space Sciences, Advisor: Jinshui Huang*

*Hefei, China*

- Develop time-dependent geodynamic models coupling early Réunion plume-lithosphere interaction ( 70-66 Ma) with an evolving double-subduction system, and quantify their respective contributions to India's acceleration, sustained rapid motion, and eventual slowdown.

## Publications

- [1] G.\* Gianni, **He, X.\***, T. W. Becker, and C. Faccenna. "Western Pacific double-subduction drives Gulf of California rifting". In: *Science* (2026). Accepted.

## Selected Research Themes

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- **Western U.S. vertical motion and Cascadia:** subduction-system dynamics, mantle flow, dynamic topography, and comparison with surface observations.
- **Mantle rheology and plate motion:** sensitivity of plate-scale deformation and surface observables to mantle viscosity structure and density heterogeneity.
- **Subduction and plume dynamics:** numerical models of single/double subduction, plume–plate interactions, and long-term tectonic evolution.

## Technical Skills

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**Programming:** Python, C, Fortran, MATLAB, shell scripting

**Geodynamic / Geophysical Software:** ASPECT, CitcomS, GPlates, GMT / PyGMT, ParaView

**Scientific Computing:** HPC workflows, MPI-based simulations, Linux, Git, numerical modeling, inverse problems, data–model comparison

**Methods:** Finite-element geodynamics, sensitivity analysis, geophysical data processing, visualization

## Awards & Honors

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|-----------------------------------------------------------|-----------|
| Graduate Fellowship, UT Austin                            | 2024–2025 |
| Guo Moruo Scholarship, USTC                               | 2024      |
| China National Scholarship (top 0.2% in China)            | 2023      |
| Yang Yongman Scholarship, USTC                            | 2023      |
| USTC Outstanding Student Scholarship, Gold Medal (top 3%) | 2022      |

## Teaching

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**University of Science and Technology of China**

*Teaching Assistant, Electromagnetism A (PHYS1004A.04.2023SP)*

Instructor: Prof. Gan Qin

**Mar. 2023–Jun. 2023**

*Hefei, China*